

TANIA CLAYTON

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EDUCATION

Master of Economics (Applied Track) | National University of Singapore, Singapore Aug 2026 – Jul 2028

- Awarded merit-based rebate by the Department of Economics.

BA (Hons.) Economics | Madras School of Economics, Chennai Aug 2023 – May 2026

- CGPA: 8.80/10
- **Relevant Modules:** Econometrics, Time Series Analysis, Applied Econometric Methods, Mathematical Methods I & II, Real Analysis, Stochastic Processes, Game Theory, Differential Equations, Python/MATLAB
- **Teaching Assistant** for Mathematical Methods II and Introduction to Python & Data Science
- Selected by the founding dean for an extra-curricular Advanced Differential Equations course

High School Diploma | St. Michael's Academy, Chennai Jun 2008 – Mar 2023

- **All India Rank 8 (98%)**, ISC Class XII Boards, **Valedictorian**

PROFESSIONAL EXPERIENCE

Quantitative Finance Intern | Janus Technology (Finance Wing), Chennai May 2026 – Present

- Designed, built, and deployed a **full-stack trading analytics platform** (Python, Flask, MySQL, AWS) computing a proprietary Banker RSI indicator across NSE, NASDAQ, and NYSE equities, extending from end-of-day to intraday data.
- Engineered a multi-factor signal suite combining technical indicators (TTM Squeeze, VWAP, Bollinger Bands, support/resistance, market-regime detection, momentum, mean reversion) with macro signals from automated news scraping and NLP-based sentiment analysis on social media data.
- Built options flow analysis, trade-logging, and execution-tracking modules producing backtested, confidence-weighted trade recommendations via a composite scoring model.
- Apply platform-generated signals to execute intraday and swing trades, managing risk against live P&L.

Student Survey Coordinator | Women in Economics and Policy, Remote May 2026 – Present

- Designed a participation strategy for the Annual Student Survey, introducing incentives that lifted response rates nearly 6x.
- Built a Power BI dashboard, handling data cleaning (Power Query), DAX measures, and visualization, to translate survey results into program recommendations.
- Co-founded the MSE chapter of WiEP, coordinating outreach and focus groups for undergraduate and graduate students.

Summer Intern | Voluntary Health Services, Chennai May 2025 – Jun 2025

- Conducted a literature review on cost-effectiveness analysis of HIV treatments and supported data collection/statistical analysis at the intersection of public health and economics.

Summer Research Intern | Energy Consortium, IIT Madras, Chennai May 2024 – Jul 2024

- Researched energy efficiency in Indian fertiliser industries under Prof. Santhosh Sahu, analysing agricultural data, emissions profiles, and emerging alternatives.

ACADEMIC PROJECTS

A Critique of Reinforcement-Learning Approaches to Collusion in Oligopoly May 2025 – Dec 2025

- Co-authored a literature review with Dr. Arun Selvan and Dr. Srikanth examining how deep reinforcement learning agents can tacitly learn collusive pricing strategies in simulated oligopoly markets, and the methodological limitations of applying these findings to real-world antitrust analysis.
- [SSRN link](#)

- Designed and ran an event study on NIFTY100 dividend announcements using CMIE Prowess data, testing abnormal returns around announcement windows to evaluate semi-strong form market efficiency.
- Applied statistical hypothesis testing and time-series methods to quantify information leakage and market reaction speed.
- Gave a talk on the topic and its tests in MSE's Research Seminar Series.

SKILLS AND ADDITIONAL INFORMATION

Technical Skills: Python, Flask, SQL (MySQL), R, STATA, MATLAB, LaTeX, AWS (Lightsail), Git/GitHub, Web Scraping, Power BI (DAX, Power Query), Excel (data analysis & dashboarding), AI Fluency and Coding, Econometrics, Regression Analysis, Time Series Analysis, Experimental design, Event Study Methodology, Stochastic Processes, NLP & Sentiment Analysis, Backtesting, Data Cleaning & Wrangling, Data Visualization, Business Intelligence Reporting, Dashboard Development

Financial Skills: Equity Research, Options Flow Analysis, Technical Analysis, Composite Signal Scoring, Intraday & Swing Trading, Risk Management, Financial Data Analysis

Languages: English (Native), Tamil (Native), Hindi (Fluent), German (A1)

Leadership & Volunteering

- **Vice President**, MSE Cultural Committee (2025–26): Led ~30 events on a budget of INR 150,000+; secured sponsors and grew social reach to 6,000+ impressions.
- **Besant Memorial Animal Dispensary:** Occasional volunteer and shelter care.
- **Nirvriti-Lit Club:** Reading sessions for children with disabilities, Vidya Sagar.